

LESSON 6.7

APPLYING SIMILAR TRIANGLES TO SOLVE REAL-WORLD PROBLEMS



LESSON INTRODUCTION

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

LEARNING TARGETS

- I can apply similar triangles to solve real-world problems.

BENCHMARK COHERENCE

BUILDING ON

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

WORKING TOWARD

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

ESSENTIAL QUESTIONS

- How can you apply what you have learned about similar triangles to solve real world problems?

LESSON NARRATIVE

In this lesson, students use their knowledge of similar triangles to approximate the height of various objects in the classroom and/or outside. This lesson uses a mirror and similar triangles to approximate the height of objects. Students have hands-on experiences with real-world applications of similar triangles.

COMPONENT SUMMARY

6.7.1 WARM-UP (~5 MIN.)

Use clues to solve puzzle

6.7.2 EXPLORATION (25 MIN.)

Explore using the mirror method to approximate height

6.7.3 YOUR TURN! (10 MIN.)

Practice using mirror method to approximate height

6.7.4 WRAP-UP (~5 MIN.)

Reflect on mastery of unit benchmarks

RESOURCES

GLOSSARY

Key Terms:

- Similar triangles
- Corresponding sides
- Constant of proportionality

Additional Terms: N/A

MATERIALS

- One yardstick or measuring tape per group
- One small mirror per group
- Calculators
- (Optional) Highlighters

ADDITIONAL PREPARATION

- (Optional) 6.7.2 Exploration can be done outside to allow for more options of objects to measure.
- Consider choosing some objects ahead of time for students to use in 6.7.2 Exploration.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Student measurements may be inaccurate.
- Students may create incorrect diagrams or label the diagrams incorrectly.

THINGS TO CONSIDER

- Consider modeling how to measure using the yardstick or measuring tape.
- Consider providing pictures along with instructions for each step of the mirror method.
- Consider that the Exploration is an opportunity to experience mathematics. Student estimations of the heights of objects may be inaccurate. Consider not focusing on the accuracy, but on the correct set up and experiencing similar triangles.
- If time is an issue, then consider assigning 6.7.3 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Take Turns

In 6.7.2 Exploration, students should take turns as they complete the activity. Each student will have the opportunity to complete the different roles required in the mirror method. Students discuss their reasoning and obtain feedback from their partner. These conversations help students build confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

In 6.7.2 Exploration, students apply their knowledge of similar triangles and discover how to use it to approximate the heights of objects. Students use real objects either in the classroom or outside to practice using this method.

DIFFERENTIATE INSTRUCTION

For 6.7.2 Exploration, consider having students work in small groups or with a partner. Make sure that each student has a role with defined responsibilities. Consider having students rotate roles for each problem. Assigning roles will hold students accountable for participating.

6.7.1 WARM-UP: CRACK THE CODE

DIFFERENTIATE INSTRUCTION

Prompt students to start with the second clue. This clue tells them which numbers are not in the puzzle. Have students list the numbers 0-9 below each of the three spots. Then, remind them to cross out the numbers as they determine those which are not possible.



6.7.1 Warm-Up: Crack the Code

- Feedback is given on four combination attempts to crack the code. Use the feedback on each attempt to crack the code.

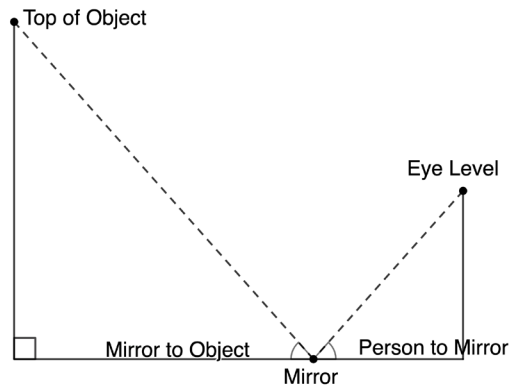
Combination Attempt	Feedback
5 6 8	One number is correct and is correctly placed.
5 8 9	Nothing is correct.
2 9 6	One number is correct, but it is in the wrong place.
4 9 1	Two numbers are correct, but they are in the wrong place.

- What is the code?
164
- Explain why your code is correct.
Feedback 2 says that the numbers 5,8, and 9 are incorrect. So referring back to feedback 1,6 has to be the number in the middle. Feedback 3 confirms that 6 is in the middle. Feedback 4 says that 4 and 1 are the correct numbers, they are just in the wrong place. The number 9 from step 4 is not used because of feedback number 2. Giving the code 164.



6.7.2 Exploration: Mirror, Mirror on the Ground

If an object is visible in a mirror positioned on the ground, then the measures of the angles of reflection from the eye to the mirror, and the mirror to the object are equal and similar triangles are created.



In this exploration, use a mirror and similar triangles to estimate the height of objects.

Materials required: 1 yardstick per group
1 mirror per group

With your partner, determine who is Partner A and who is Partner B. Then, work together to follow the steps.

6.7.2 EXPLORATION: MIRROR, MIRROR ON THE GROUND

QUESTIONING

Consider having students work in small groups or with a partner. Make sure that each student has a role with defined responsibilities. Consider having students rotate roles for each problem. Assigning roles will hold students accountable for participating.

DIFFERENTIATE INSTRUCTION

Consider having students complete this activity outdoors, since there are taller objects to measure.

If working in the classroom, some suggested objects students could approximate the height of include desk, chair, person, window, etc.

TEACHER MOVES

While students are working, consider collecting observational data using a rubric. Use the observations to inform what differentiation or small-group instruction is needed. Specifically focus on what students have noted in their reflections as well as your observations. Use this information to utilize small-group instruction during Your Turn!.

6.7.2 EXPLORATION: MIRROR, MIRROR ON THE GROUND (CONTINUED)

ENRICHMENT

How does this strategy use similar triangles?

DIFFERENTIATE INSTRUCTION

For step 2, students can use any distance and this strategy will still work. However, having students use multiples of 10 would make calculations easier. Consider reviewing strategies for measuring the distance between a person and the object if the distance is longer than one yardstick (i.e. by marking where the yardstick ends, then bringing the yardstick to measure from there).

QUESTIONING

Consider having one mystery object for which you give the students the actual height and students have to find the object with that height.

TEACHER MOVES

This strategy works best with taller objects, so encourage students to choose tall objects to measure.

Step 1: Partner B measures the distance from the floor to Partner A's eyes. Label the above diagram with the measurement.

Height of eye level: Answers will vary inches

Step 2: Partner A places the mirror on the floor between themselves and the object to be measured, then measures the distance from the base of the object to the center of the mirror. Label the measurement on the diagram.

Step 3: Partner A starts at the mirror and walks backwards from the mirror and the object, stopping when the top of the object can be seen in the middle of the mirror.

Step 4: Partner A stays in place and Partner B measures the distance from Partner A's feet to the middle of the mirror. Label the measurement on the diagram.

Step 5: Use your knowledge of similar triangles to estimate the height of the object.

Repeat steps 2-5, alternating roles between Partner A and Partner B. Record your results in the table.

Answers will vary.

Object	Distance from Partner to Mirror	Distance from Object to Mirror	Work to Determine Approximate Height

6.7.2 EXPLORATION: MIRROR, MIRROR ON THE GROUND (CONTINUED)

TEACHER MOVES

Take Turns Students should take turns as they complete the activity. Each student will have the opportunity to complete the different roles required in the mirror method. Students discuss their reasoning and obtain feedback from their partner. These conversations help students build confidence and understanding.

DIFFERENTIATE INSTRUCTION

Students may be confused about where to place the mirror. Consider giving students an exact distance rather than having them choose the distance.

ENRICHMENT

- Why is the height that we get an approximation and not an exact height?
- In what situations might it be useful to have an approximate height?

6.7.3 YOUR TURN!

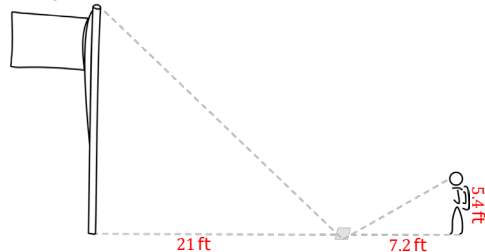
DIFFERENTIATE INSTRUCTION

Consider having students highlight and label important information in the problem before labeling the diagram. Allow students to use calculators to find the constant of proportionality.



6.7.3 Your Turn!

1. Mrs. Robinson assigned her class a project to find the height of a flagpole. The students could not measure the height, so they used a mirror to determine the height of the flagpole. Tayveon placed a mirror on the ground 21 feet (ft.) from the base of the flagpole and backed up until the top of the pole was centered in the mirror.



Tayveon is 5.4 ft. tall and is standing 7.2 ft. from the mirror.

- a. Label the diagram with the given measurements.
- b. How tall is the flagpole?

$$\frac{\text{Base from the flag pole to the mirror}}{\text{Mirror to Tayveon}} = \frac{21 \text{ ft.}}{7.2 \text{ ft.}} = 2.91\bar{6}$$

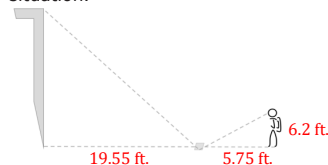
2.91 $\bar{6}$ is the constant of proportionality. The flagpole is Tayveon's height multiplied by the constant of proportionality; $5.4 \times 2.91\bar{6} = 15.75$ ft. The flagpole is 15.75 ft. tall.

DIFFERENTIATE INSTRUCTION

Students may not be sure how to represent the situation. Consider providing struggling students with the diagram or explaining that the mirror is a small mirror placed horizontally on the ground.

2. A rock climber wants to know the height of a cliff. She places a mirror 19.55 feet (ft.) from the base of the cliff, then backs up from the mirror until she can see the top of the cliff in the center of the mirror. At that point, her eye level is 6.2 ft. above the ground and she is standing $5\frac{3}{4}$ ft. from the mirror.

- a. Draw and label a diagram that represents this situation.



- b. Find the height of the cliff to the nearest tenth of a foot.

$$\frac{\text{Base of cliff to mirror}}{\text{Rock climber to mirror}} = \frac{19.55 \text{ ft.}}{5.75 \text{ ft.}} = 3.4 \quad 6.2 \times 3.4 = 21.08 \text{ ft}$$

The constant of proportionality multiplied by the climber's height is 21.08 ft.



6.7.4 Wrap-Up: Reflection

1. In this unit, there were three benchmarks. For each of the benchmarks, write one thing you feel confident you know and one thing you are still unsure about.
- Answers will vary.*

Benchmark	One thing I am confident I know is ...	One thing I am still unsure about is ...
8.AR.3.1 Determine if a linear relationship is also a proportional relationship.		
8.AR.3.2 Given a table, graph or written description of a linear relationship, determine the slope.		
8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.		

6.7.4 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider allowing students to use speech-to-text software or an audio recorder (i.e. Flipgrid) if they struggle expressing themselves in writing.